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Floating marine platform

Abstract

A floating marine platform is provided including a central column, at least three peripheral columns circumferentially around the central column, radially extending beams from the central column that connect the peripheral columns with the central column, and structural members spanning between each adjacent pair of peripheral columns. The structural members are pre-tensioned.

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Background/Summary

BACKGROUND
(1) The invention relates to a floating marine platform.
SUMMARY OF THE INVENTION
(2) Growing concerns related to climate change gave impetus to renewable power generation, including a potential large market for wind power generation offshore. Existing large-scale developments are taking place in relatively shallow waters with structures fixed to the seabed. However, these technologies restrict such development to areas that are generally relatively close to the shore, with a potential visual impact, and with favorable seabed conditions. Additionally, fixed wind turbines tend to require heavy work offshore, with associated risk to the workers and financial risks due to adverse weather and the need for very large offshore construction vessels.
(3) The ability to provide floating foundations for wind turbines can substantially increase the areas available for offshore wind farm development, by locating these units in deeper water, further away from the shore, where visual impacts tend to reduce and wind speed is generally higher and less turbulent. In addition, these may require less work to be conducted offshore, as the turbine can be fully assembled at port and towed to site.
(4) Several innovative demonstration projects have been completed successfully in recent years to demonstrate the feasibility of installing commercial-size wind turbines on floating foundations. However, large-scale development has not happened yet due to the high cost of the structures already deployed, and the difficulty to mass-produce the floaters with existing shipyard or civil works facilities. There is a need for reducing the size and cost of the hulls proposed for floating wind turbine foundations, while accommodating very large wind turbines that are planned to be used for commercial scale development, with a rotor diameter close to 200 meters.
(5) It is an object of the present invention to provide a floating marine platform that is more economical to produce while providing stability that is substantially equivalent to or higher than that of existing floating marine platforms, in particular of floating foundations for offshore wind turbines.
(6) According to a first aspect, the invention provides a floating marine platform comprising a central column, at least three peripheral columns circumferentially around the central column, radially extending beams from the central column that connect the peripheral columns with the central column, and structural members spanning between each adjacent pair of peripheral columns, wherein the structural members are pre-tensioned.
(7) The pre-tension of the structural members induce a counter pressure to the beams that biases the beams in their elongate direction towards the central column. Due to the pre-tension the structural

members remain in tension, and the beams remain in compression during the load cycles as exerted on the platform by, for example, waves. This configuration of the structural members provides stiffness to the beams in the horizontal plane and so reduces the moment forces in the beams. As the structural members and the beams can be designed to mainly resist tension forces and compression forces respectively, the structural members and the beams can be executed more slender and lighter as compared to conventional floating marine platforms. As the floating marine platforms comprises similar or identical central and peripheral columns the stability will largely be the same as that of the existing floating marine platforms.
(8) In an embodiment the radially extending beams comprise a top beam and a bottom beam that extend parallel to each other. In further embodiments the beams have a circular cross section and/or an I-shaped cross section. The beams are configured as an outrigger in a truss configuration that is composed of conventional steel sections. This is a straightforward steel structure which can be produced in an economical way.
(9) In some embodiments the central column and/or the peripheral columns extend vertically.
(10) In an embodiment the peripheral columns and the structural members form a generally triangular shape, preferably an equilateral triangular shape. The triangular configuration provides a form stable arrangement of the peripheral columns and the structural members that is straightforward to produce.
(11) In an embodiment the structural members comprise or are formed with a steel tube to deliver a high pre-tension in the structural members and the beams.
(12) In a lightweight alternative embodiment the structural members comprise or are formed with a steel or aramid fiber cable.
(13) In an embodiment the peripheral columns comprise a connector having a passage for one end of the structural member, wherein the structural member comprises a tension head at the end that is received in the connector and the structural member extends through the passage as from the tension head. The tension head can be hooked into the connector where after it can be pre-tensioned.
(14) In an embodiment thereof the peripheral columns comprise one or more shims or shim plates between the connector and the tension head to maintain the pre-tension in the structural members after it has been built.
(15) In an embodiment at least one of the beams serves as a gangway to provide access between the central column and at least one of the peripheral columns.
(16) In some embodiments the structural members are located at an elevation below that of the radially extending beams and/or the structural members are located at an elevation above that of the radially extending beams to further assure that compression in specific parts of the beams or steel sections thereof is maintained.
(17) In an embodiment the beams connect to the columns using bolted connections. This is a straightforward connection method which can be produced in an economically favorable way.
(18) The structural members that extend in a common horizontal plane preferably have the same pre-tension to provide symmetry around or along their spanning. The pre-tension in a first set of bottom structural members and a second set of top structural members may differ from each other in order to differentiate bias in the bottom and top beams when required.
(19) In an embodiment the structural members are pre-tensioned by inducing a pre-tension stroke thereto that is between 0.04% and 0.07% of the length of the structural member, preferably 0.05% of the length of the structural member. In this way the permanent deformation of the structural members will remain very small, as they operate in the elastic range. The structural members will therefore remain in tension during the lifetime of the floating marine platform. Due to the pre-tension the structural members remain in tension at all times, except during the largest waves of the most powerful storms, whereby they may occasionally get slack for brief periods, such as a few seconds of the wave cycle at most.

(20) In an embodiment the beams are biased in their elongated direction towards the central column, preferably the beams that extend in a common horizontal plane have the same bias, providing stiffness to the beams in the horizontal plane and so reducing the moment forces in the beams. The bias in the top and bottom beams of the radially extending beams may differ from each other in order to further improve the stiffness of the radially extending beams.
(21) In an embodiment the peripheral columns comprise a buoyancy air chamber. The air chambers provide buoyancy to the floating marine platform at the peripheral columns. As the peripheral columns are located at a distance from the central column, the additional buoyancy provides stability to the floating marine platform.
(22) In an embodiment the peripheral columns comprise a base and the buoyancy air chamber is open to the sea at the base. In this way the buoyancy of the peripheral columns can be adjusted by adjusting the air level in the air chamber and therewith forcing water in or out the air chamber via the base.
(23) In an embodiment the floating marine platform further comprises a motion control system comprising a high-pressure air tank for discharging air in the open buoyancy air chambers, which controls airflow in and out of the open buoyancy air chambers using actuated valves and is controlled by a computer system that is coupled with motion sensors. The motion control system can adjust the air level in the open buoyancy air chambers to provide a moment opposing that caused by the aerodynamic forces on the floating marine platform, and therewith stabilize the floating marine platform.
(24) In an embodiment thereof the high-pressure air tank is formed by the airtight inside volume of one of the beams. By using the beams as the air tank, no additional air tank needs to be provided, which saves space on the floating marine platform.
(25) In a further embodiment the motion control system comprises an air compressor that is configured to fill the high-pressure air tank. In an embodiment thereof the motion control system comprises an inlet valve that is configured to control the filling of the high-pressure air tank by the air compressor.
(26) In an embodiment the motion control system comprises for each open buoyancy air chamber an outlet valve that is configured to control the discharging of air from the high-pressure air tank into the corresponding open buoyancy air chamber.
(27) In an embodiment the motion control system comprises for each open buoyancy air chamber a release valve that connects the open buoyancy air chamber to the atmosphere and that is configured to control the releasing of air from the open buoyancy air chamber to the atmosphere. The air compressor, the inlet valve, the outlet valve and the release valve are conventional parts that can be implemented and installed in an economical favorable way.
(28) According to a second aspect, the invention provides a floating marine platform comprising a central column, at least three peripheral columns circumferentially around the central column, radially extending beams from the central column that connect the peripheral columns with the central column, wherein the peripheral columns comprise a buoyancy air chamber, and wherein the marine platform comprises a motion control system comprising a high-pressure air tank discharging air in the buoyancy air chamber, which controls airflow in and out of the buoyancy air chambers using actuated valves and is controlled by a computer system that is coupled with motion sensors.
(29) The floating marine platform according to the second aspect and its embodiments relate to the floating marine platform according to any one of the aforementioned embodiments and thus have the same technical advantages, which will not be repeated hereafter.
(30) The various aspects and features described and shown in the specification can be applied, individually, wherever possible. These individual aspects, in particular the aspects and features described in the attached dependent claims, can be made subject of divisional patent applications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be elucidated on the basis of an exemplary embodiment shown in the attached drawings, in which:

(2) FIGS. **1**A, **1**B and **1**C are an isometric view, a side view and a top view of a floating marine platform according to a first embodiment of the invention, that supports a wind turbine;

(3) FIGS. **2**A and **2**B are close ups of the floating marine platform according to FIGS. **1**A and **1**B respectively, with partial cut-outs to show some internal parts thereof;

(4) FIG. **3** is a detailed view of a connection between a tendon and a skirt of the floating marine platform;

(5) FIGS. **4**A and **4**B are an isometric view and a side view of a floating marine platform according to a second embodiment of the invention, that supports a wind turbine;

(6) FIG. **5** shows a top section view of a peripheral column of a floating marine platform according to a third embodiment of the invention; and

(7) FIG. **6** is an isometric view of a floating marine platform according to a fourth embodiment of the invention, that supports data measurement and acquisition equipment.

DETAILED DESCRIPTION OF THE INVENTION

(8) FIGS. **1**A-**1**C and **2**A-**2**B show a floating marine platform **1** according to a first embodiment of the invention. The marine platform **1** supports in this example a wind turbine **300** to form a floating wind turbine **10**. The wind turbine **300** has a vertical tower **301** and a nacelle **302** on top of the tower **301** having an internal generator that is driven by a wind turbine rotor **303**. The wind turbine rotor **303** has a hub **304** that is connected to the generator, and in this example three blades **305** radiating from the hub **304**. The wind turbine **300** is capable of producing more than 1 MW of electrical power, currently reaching about 10 to 12 MW. The bottom diameter of the tower **301** may be between 5 meter and 10 meter for a +10 MW wind turbine. The three blades **305** may be more than 100 meters long each. An example is the 12 MW Haliade X turbine from General Electrics. Other turbine designs, such as vertical axis wind turbines can also be supported by the floating marine platform **1**.

(9) As best shown in FIG. **2**B, the marine platform **1** comprises a vertical cylindrical central column **2** with a varying diameter that is approximately equal to the bottom diameter of the tower **301** at the top of the central column **2** and that increases in diameter towards the bottom or keel of the central column **2**. The central column **2** has a circumferential wall **3** and a bottom wall **4** that define an internal chamber **5**. The central column **2** is made of steel or concrete.

(10) The marine platform **1** comprises in this example three vertical stabilizing or peripheral columns **20***a*, **20***b*, **20***c* that are disposed radially every 120 degrees around the central column **2**. As best shown in FIG. **2**B, the peripheral columns **20***a*, **20***b*, **20***c* each comprise a steel first cylindrical body **21** having a circumferential wall **24**, a top wall **25**, and an internal steel watertight flat **26** that extends in this embodiment parallel with the top wall **25** and at substantially half the height of the peripheral columns **20***a*, **20***b*, **20***c*. The watertight flat **26** defines an internal closed off air chamber **71** at the upper side, and an internal open air chamber **72** at the lower side. The open air chamber **72** is open at the base or bottom edge of the circumferential wall **24** to be filled at least in part with water. The peripheral columns **20***a*, **20***b*, **20***c* comprise horizontally extending steel skirts **27** around the bottom edge of the circumferential wall **24**.

(11) The marine platform **1** comprises three outriggers **40***a*, **40***b*, **40***c* having the same radial length. The outriggers **40***a*, **40***b*, **40***c* are composed of tubular structural members and that are arranged in a truss configuration. The outriggers **40***a*, **40***b*, **40***c* connect the central column **2** to the peripheral columns **20***a*, **20***b*, **20***c*. As best shown in FIG. **2**B, the outriggers **40***a*, **40***b*, **40***c* comprise a substantially horizontal top beam **45** and a substantially horizontal bottom beam **46** that extend

parallel to each other and that are interconnected with cross members or diagonal bracings **47** if these are required. The top beam **45** and the bottom beam **46** are steel hollow cylindrical pipes, and the bracings **47** are steel hollow cylindrical pipes with a smaller diameter. The outriggers **40***a*, **40***b*, **40***c* are connected with the central column **2** and the peripheral columns **20***a*, **20***b*, **20***c* by welding, or by means of flanges that are bolted to each other to form bolted connections.
(12) The central column **2** includes a vertical cylindrical top section **6** with a constant diameter to which the top beams **45** are connected. The vertical cylindrical section **6** merges downwardly via a flared section or conically widening section **7** into a vertical cylindrical bottom section **8** with a constant diameter to which the bottom beams **46** are connected. The central column **2** may be provided with a non-shown footing below the bottom section **8** with a larger diameter that provides additional volume. When the footing is filled with air, it helps to support the weight of the wind turbine **300**. When the footing is filled with water, it helps to provide stability to the floating wind turbine **10**.
(13) The marine platform **1** comprises six pre-tensioned slender structural members or tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* having the same length that interconnect the peripheral columns **20***a*, **20***b*, **20***c*. In this example three tendons **60***a*, **60***b*, **60***c* interconnect the peripheral columns **20***a*, **20***b*, **20***c* at the skirts **27** and three tendons **61***a*, **61***b*, **61***c* interconnect the peripheral columns **20***a*, **20***b*, **20***c* near the top wall **25** thereof.
(14) A detail of a connection between a tendon **61***b* and a column **20***b*, in particular at the skirt **27** thereof, is shown in FIG. **3**. The skirt **27** comprises an annular steel base plate **28**, and a steel edge plate **29** connected thereto that defines a passage **37** towards a tendon connector **30** for the end of the connected tendon **61***b*. The tendon connector **30** comprises two pairs of parallel steel first support plates **31** that extend radially on both sides of the passage **37** and that are connected to the edge plate **29** and the base plate **28**, two steel second support plates **32** parallel to the first support plates **31** that are connected with the edge plate **29**, the base plate **28** and the circumferential wall **24**, and two stiffeners **33** made of steel plate that extend transverse to and that interconnect the first support plates **31** and second support plates **32**. The same or a similar connector **30** is provided at the top of the peripheral column **20***b*. It is to be understood that the above described tendon connector **30** is just one example of several possible embodiments. For instance, the tendon connector **30** may not be part of the skirt **27** but may instead be integrated into or attached to a column **20***b*.
(15) The tendon **61***b* comprises a hollow cylindrical steel pipe **62** and a tension head **63** at the end of the pipe **62**. The tension head **63** comprises two supports **64** projecting from the pipe **62** at opposite sides thereof. The steel pipe **62** of the tendon **61***b* extends at the end through the passage **37**, and the tension head **63** is received in the connector **30** where it hooks behind the first support plates **31**. The tendon **61***b* is pre-tensioned by means of temporarily installed hydraulic cylinders **35** that are at the end of the tendon **61***b* positioned between the third support plates **33** of the connector **30** and the supports **64** of the tendon **61**. The hydraulic cylinders **35** push the tension head **63** away from the first support plates **31**, whereafter the gap is permanently filled up with steel shim plates **36**. After pre-tensioning of the tendon **61***b* the hydraulic cylinders **35** are removed.
(16) Each tendon **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* comprises at least one tension head **63** at one end thereof, and each peripheral column **20***a*, **20***b*, **20***c* comprises at least one corresponding tendon connector **30** so that each tendon **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* can be pre-tensioned. The tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* may comprise two tension heads **63**, one at each end thereof, the peripheral columns **20***a*, **20***b*, **20***c* may comprise two tendon connectors **30** that correspond to the respective tension heads **63**. The tendon **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* could also, at the end opposite to the tension head **63**, comprise a forged axi-symetrical head that corresponds to a connector at the other peripheral column **20***a*, **20***b*, **20***c*. The pre-tension of the tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* causes that in at least all the bottom beams **46**, but in this example in all beams **45**, **46**, a counter pressure force is induced that biases the bottom beams **46**, and in this example

also the top beams **45**, in their elongated direction towards the central column **2**. Due to the pre-tension in the tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* these structural members remain in tension at all times, except during the largest waves of the most powerful storms, whereby they may occasionally get slack for brief periods, such as a few seconds of the wave cycle at most. The tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* provide stiffness to the outriggers **40***a*, **40***b*, **40***c* in the horizontal plane.

(17) The marine platform **1** is provided with a gangway **11** that extends around the bottom of the tower **301** and above one or more of the top beams **45** towards the top side of the peripheral columns **20***a*, **20***b*, **20***c*. Alternatively the top beams **45** are for example I-beams or H-beams which may be used by technicians as gangway for access between the central column **2** and the peripheral columns **20***a*, **20***b*, **20***c*.

(18) The marine platform **1** comprises a motion control system of which some components are schematically shown in FIG. **2**B. The motion control system is configured to adjust the air level in the open air chambers **72** in order to stabilize the marine platform **1** and to provide a moment opposing that caused by the aerodynamic forces on the wind turbine rotor **303**. The motion control system comprises an air compressor **75** that is located in the central column **2** and that is powered by the turbine electrical system. The air compressor **75** is connected to air distribution pipes **76** that connect to inlet valves **77**. The inlet valves **77** communicate with an airtight compartment, a high-pressure air chamber or high-pressure air tank **82** which is the inside of the bottom beams **46** or that is formed by the entire airtight inside volume thereof. The air tank **82** is filled by the air compressor **75** with high-pressure air between 7 and 12 bars. The air compressor **75** may be configured to automatically start filling the air tanks **82** when the pressure drops below a preset value. The power to run the air compressor **75** (approx. 100 kW or less) may be provided by the turbine systems. If there is insufficient wind at the site and the wind turbine **300** is not producing electrical power, the grid may be used to provide the power.

(19) The motion control system comprises for each open air chamber **72** an automatic outlet valve **78** that can be controlled to release the high pressure air from the air tank **82** into an air outlet **81** at the bottom of the open air chamber **72**. The amount of air being released can therefore be controlled precisely with the opening of the automatic outlet valve **78**. An air nozzle or air outlet **81** may be added downstream of the automatic outlet valve **78** to direct the air flow towards the base of the open air chamber **72**, which will produce a dynamic lift force. Alternatively the air flow is directed towards the top of the open air chamber **72**, the water escaping from the open air chamber **72** through the base the causes a dynamic lift force. The motion control system comprises in each peripheral column **20***a*, **20***b*, **20***c* a vent line **79** with an automatic release valve **80** that connects the top of the open air chamber **72** to the atmosphere above the top of the peripheral column **20***a*, **20***b*, **20***c*. The automatic release valve **80** can be controlled to open and close the vent line **79**. The motion control system comprises 6 degrees of freedom instruments to monitor translations and rotations of the marine platform **1** in three perpendicular directions.

(20) The motion of the marine platform **1** is monitored by the motion control system. When there is a change in wind speed or direction, the floating wind turbine **10** heels. When there is a mean change of heel, the automatic outlet valve **78** between one or more high-pressure tanks **82** and bottom chambers **72** of the peripheral columns **20***a*, **20***b*, **20***c* will open to let air in the open air chambers **72** connected to the sea, if air quantity needs to be increased, and/or one or more automatic release valves **80** controlling the atmospheric vents **79** will open if air quantity needs to be decreased. Improved rotor tilt can be achieved with the motion control system to enhance power production.

(21) Similarly, if a shutdown of the wind turbine **300** is triggered, including emergency shutdown due to loss of grid power or any other issue causing such turbine response, the air in the open air chambers **72** of the peripheral columns **20***a*, **20***b*, **20***c* will be adjusted by opening the corresponding automatic outlet valve **78** and release valve **80**. This will reduce the maximum

inclination of the marine platform **1** expected to occur due to such event. The marine platform **1** will then be returned to even-keel condition, until the wind turbine **300** is ready to start.

(22) If the sea-state is high, the automatic outlet valve **78** and release valve **80** may be opened based on the timing of the motion response to reduce the wave-induced response. This will increase the capacity of the floating wind turbine **10** to operate efficiently in heavy seas.

(23) The motion control system can ensure that the tower **301** of the wind turbine **300** remains at an optimal angle for production of power in the wind farm. This is advantageous as most large-size commercial wind turbines are three-bladed upwind turbines having the rotor **303** tilted upward looking toward the direction where the wind is coming from in order to prevent collision of the blades **305** with the tower bottom due to their deflection caused by aerodynamic loads. By operating the motion control system, it is prevented that the tilt angle increases significantly, which would cause a reduction of the wind load and produced electrical power.

(24) In the shown embodiment, a first set of pre-tensioned tendons **61***a*, **61***b*, **61***c* interconnecting the peripheral columns **20***a*, **20***b*, **20***c* is provided near the top of the peripheral columns **20***a*, **20***b*, **20***c* and/or at an elevation that is above the outriggers **40***a*, **40***b*, **40***c*, and a second set of the pre-tensioned tendons **60***a*, **60***b*, **60***c* is provided near the bottom at the skirts **27** and/or at an elevation that is below the outriggers **40***a*, **40***b*, **40***c*. Alternatively the pre-tensioned tendons **61***a*, **61***b*, **61***c* interconnecting the peripheral columns **20***a*, **20***b*, **20***c* may be provided only near the top of the peripheral columns **20***a*, **20***b*, **20***c* and/or at an elevation that is above the outriggers **40***a*, **40***b*, **40***c*. In still alternative embodiments the pre-tensioned tendons **60***a*, **60***b*, **60***c* interconnecting the peripheral columns **20***a*, **20***b*, **20***c* may be provided only near the bottom at the skirts **27** and/or at an elevation that is below the outriggers **40***a*, **40***b*, **40***c*. In yet other embodiments, the pre-tensioned tendons **60***a*, **60***b*, **60***c* interconnecting the peripheral columns **20***a*, **20***b*, **20***c* may be provided only near the center of the peripheral columns **20***a*, **20***b*, **20***c* and at an elevation that is about the same as that of the outriggers **40***a*, **40***b*, **40***c*.

(25) As shown in FIG. **2**B, the cylindrical top section **6** of the central column **2** matches that of the bottom diameter of the tower **301**, whereby the cylindrical top section **6** has a top diameter D**1** of 5-10 meters. The central column **2** may then flare outwardly to reach a bottom diameter D**2** at the cylindrical bottom section **8** of up to 20 meters. The central column **2** and the peripheral columns **20***a*, **20***b*, **20***c* typically have a total height H**1** of 20-30 meters, in this example about 24 meters. The peripheral columns **20***a*, **20***b*, **20***c* have a diameter D**3** between 6-12 meters. The watertight flat **26** extends at about half of the total height H**1**, whereby the open air chamber **72** has a chamber height H**2** of 5-15 meters, in this example about 9 meters. At sea, the water enters the open air chamber **72** up to a water height H**3** of 3-9 meters, in this example about 6 meters, that is lower than the chamber height H**2**, whereby there is always compressed air present in the open air chamber **72**. The top of the central column **2** and the peripheral columns **20***a*, **20***b*, **20***c* may be up to 10-15 meters above the mean water level, and the draft may vary between 10-15 meters.

(26) The steel components of the marine platform **1** are formed from S355, marine grade mild carbon steel. Higher strength steel may also be used for some components.

(27) As shown in FIG. **1**C the tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* each have a length L**1** of 60-90 meters, in this example about 73 meters. The tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c* have a nominal diameter D**4** of 300-600 millimeters meters and a wall thickness of 15-30 millimeters. In some embodiments, higher strength steel may be used to apply a wall thickness of 5-18 millimeters. The length of the pre-tension stroke as induced by the hydraulic cylinders **35** is 0.04-0.07% of the length L of the tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c*, in this example 0.05% of the length L of each tendon **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c*. This induces a pretention force in the tendons **20***a*, **20***b*, **20***c* of 200-300 tons, in this example about 250 tons. The pretension forces in the upper tendons **60***a*, **60***b*, **60***c* are about the same, and the pretension forces in the lower tendons **61***a*, **61***b*, **61***c* are about the same, and the counter pressure that is induced in the beams **45**, **46**, are about the same, giving a force symmetry around the spanning of the tendons **60***a*, **60***b*, **60***c*, **61***a*, **61***b*, **61***c*.

(28) The floating wind turbine **310** is kept on station with mooring lines connected to the bottom of each peripheral column **20***a*, **20***b*, **20***c* through a mooring line connector that can be closed to keep the marine platform **1** on site while the wind turbine **300** is producing electrical power, or that can be open if the floating wind turbine **310** needs to be towed back to shore for maintenance or decommissioning.
(29) In the described first embodiment the flat **26** divides the first cylindrical body **21** vertically into the closed off air chamber **71** and the open air chamber **72**. Alternatively, as schematically shown in FIG. **2**B with broken lines, the first cylindrical body **21** is horizontally divided by a vertical plate **93** into the closed air chamber **71** and the open air chamber **72**, wherein the closed air chamber **71** is closed off by a horizontal keel plate **94** that extends between the bottom edge of the circumferential wall **24** and the bottom edge of the vertical plate **93**. The closed air chamber **71** is located closest to the respective outrigger **40***a*, **40***b*, **40***c*, and the air outlet **81** is located at the bottom of the open air chamber **72**.
(30) FIGS. **4**A and **4**B show a floating marine platform **101** according to a second embodiment of the invention. The marine platform **101** supports the wind turbine **300** to form a floating wind turbine **110**. The features of the floating marine platform **101** that correspond with the floating marine platform **1** according to the first embodiment are provided with the same reference numbers, and hereafter only the deviating features are described.
(31) The three peripheral columns **20***a*, **20***b*, **20***c* of the marine platform **101** comprise a steel second cylindrical body **22** adjacent to the first cylindrical body **21** at the side radially opposite to the respective outriggers **40***a*, **40***b*, **40***c*. The skirts **27** extend around the joint cylindrical bodies **21**, **22**. The cylindrical bodies **21**, **22** may both comprise the closed air chamber **71** and the open air chamber **72** like in the first embodiment, or one of the cylindrical bodies **21**, **22**, having the mooring line connector **99** again at the distal side, has the open air chamber **71** with the air outlet **81**, while the other forms the closed air chamber **71**.
(32) FIG. **5** shows a detail of a floating marine platform **201** according to a third embodiment of the invention. The marine platform **201** supports the wind turbine **300** to form a floating wind turbine **210**. The features of the floating marine platform **201** that correspond with the floating marine platforms **1**, **101** according to the first and second embodiment are provided with the same reference numbers, and hereafter only the deviating features are described.
(33) The three peripheral columns **20***a*, **20***b*, **20***c* of the marine platform **201** comprise a steel second cylindrical body **22** and a steel third cylindrical body **23** adjacent to each other, and both adjacent to the first cylindrical body **21** at the side radially opposite to the respective outriggers **40***a*, **40***b*, **40***c*. The skirts **27** extend around the joint cylindrical bodies **21**, **22**, **23**. The cylindrical bodies **21**, **22** may all comprise the closed air chamber **71** and the open air chamber **72** like in the first embodiment, or one of the cylindrical bodies **21**, **22** has the open air chamber **71** with the air outlet **81**, while the other forms the closed air chamber **71**. In this embodiment, the first cylindrical body **21** and the third cylindrical body **23** are closed off at the bottom by the keel plate **94** to form the closed air chambers **71**, while the second cylindrical body **22** is open to form the open air chamber **72** with the air outlet **81** at the bottom.
(34) FIG. **6** shows a floating marine platform **401** according to a fourth embodiment of the invention. The marine platform **401** supports in this example data measurement and acquisition equipment **412** to form an ocean monitoring platform. The ocean monitoring platform **410** may for instance be configured to perform measurements and data acquisition of metocean (wind, wave, current), chemistry (corrosion, marine growth), environmental (marine mammals, bird migration), biodiversity (juvenile and pelagic fish density) and ocean farming (shell fish and algae's growth rates, nutrient density), to prepare offshore sites for a comprehensive development.
(35) The marine platform **401** comprises a vertical cylindrical central column **402** with a constant diameter. The central column **402** has a circumferential wall **403**, a bottom wall **404**, and a top wall **409** that define an internal chamber **405** that is the main buoyancy chamber of the marine platform

401 and that houses most of the equipment **412**. The central column **402** is made of steel. The central column **402** may be provided with a footing **428** below the bottom wall **404** with a larger diameter that dampens wave induced motion of the marine platform **401**. The footing **428** may provide additional volume to the central column **402** so that, when the footing is filled with air, it helps to support the weight of the equipment **412**, and when the footing is filled with water, it helps to provide stability to the ocean monitoring platform **410**.

(36) The marine platform **401** comprises in this example three vertical cylindrical stabilizing or peripheral columns **420***a*, **420***b*, **420***c* that are disposed radially every 120 degrees around the central column **402**. The peripheral columns **420***a*, **420***b*, **420***c* each comprise a steel first cylindrical body **421** having a steel circumferential wall **424**, a top wall **425**, and a bottom wall **427** that in this example horizontally extends from the bottom edge of the circumferential wall **424** to form skirts **427**. The steel circumferential wall **424**, the top wall **425**, and the bottom wall **427** define an internal closed off air chamber **471**.

(37) The marine platform **401** comprises three outriggers **440***a*, **440***b*, **440***c* having the same radial length and that are composed of structural parts. The outriggers **440***a*, **440***b*, **440***c* connect the central column **402** to the peripheral columns **420***a*, **420***b*, **420***c*. The outriggers **440***a*, **440***b*, **440***c* comprise a substantially horizontal top beam **445** and a substantially horizontal bottom beam **446** that extend parallel to each other. The top beams **445** are steel I-beams or H-beams and the lower beams **446** are steel hollow cylindrical pipes. The outriggers **440***a*, **440***b*, **440***c* are connected with the central column **402** and the peripheral columns **420***a*, **420***b*, **420***c* by welding, or by means of flanges that are bolted to each other to form bolted connections.

(38) The marine platform **401** comprises three pre-tensioned slender structural members or tendons **460***a*, **460***b*, **460***c* interconnecting the peripheral columns **420***a*, **420***b*, **420***c* at the skirts **427** of the peripheral columns **420***a*, **420***b*, **420***c*, and three pre-tensioned slender structural members or tendons **461***a*, **461***b*, **461***c*. The tendons **460***a*, **460***b*, **460***c*, **461***a*, **461***b*, **461***c* are in this example embodied as steel or aramid fiber cables and may be pre-tensioned in a similar fashion as explained above for the marine platform **1** of FIG. **1**A-**1**C.

(39) In the shown embodiment, a first set of pre-tensioned tendons **461***a*, **461***b*, **461***c* having the same length interconnect the peripheral columns **420***a*, **420***b*, **420***c* is provided near the top of the peripheral columns **420***a*, **420***b*, **420***c* and/or at an elevation that is above the outriggers **440***a*, **440***b*, **440***c*, and a second set of pre-tensioned tendons **460***a*, **460***b*, **460***c* having the same length interconnect the peripheral columns **420***a*, **420***b*, **420***c* is provided near the bottom at the skirts **427** and/or at an elevation that is below the outriggers **440***a*, **440***b*, **440***c*. Alternatively the pre-tensioned tendons **460***a*, **460***b*, **460***c* interconnecting the peripheral columns **420***a*, **420***b*, **420***c* may be provided only near the bottom at the skirts **427** and/or at an elevation that is below the outriggers **440***a*, **440***b*, **440***c*. In still alternative embodiments the pre-tensioned tendons **461***a*, **461***b*, **461***c* interconnecting the peripheral columns **420***a*, **420***b*, **420***c* may be provided only near the top of the peripheral columns **420***a*, **420***b*, **420***c* and/or at an elevation that is above the outriggers **440***a*, **440***b*, **440***c*. In yet other embodiments, the pre-tensioned tendons **460***a*, **460***b*, **460***c* **461***a*, **461***b*, **461***c* interconnecting the peripheral columns **420***a*, **420***b*, **420***c* may be provided only near the center of the peripheral columns **420***a*, **420***b*, **420***c* and at an elevation that is about the same as that of the outriggers **440***a*, **440***b*, **440***c*.

(40) The marine platform **401** is provided with a gangway **411** on the top wall **409** of the central column **402** and above the top beams **445** towards the top side of the peripheral columns **420***a*, **420***b*, **420***c* which may be used by technicians for access between the central column **402** and the peripheral columns **420***a*, **420***b*, **420***c*.

(41) The central column **402** has a diameter of 1-3 meters, in this example about 2 meters. The central column **402** and the peripheral columns **420***a*, **420***b*, **420***c* typically have a total height of 5-15 meters, in this example about 10 meters. The peripheral columns **420***a*, **420***b*, **420***c* have a diameter between 0.5-1.5 meters, in this example about 0.8 meter. The top of the central column

402 and the peripheral columns **20***a*, **20***b*, **20***c* may be up to 5-9 meters above the mean water level, and the draft may vary between 2-6 meters. The tendons **420***a*, **420***b*, **420***c* each have a length L**2** of 10-20 meters, in this example about 15 meters.
(42) The steel components of the marine platform **1** are formed from S355, marine grade mild carbon steel.
(43) The marine platform **401** is kept on station with at least one mooring line connected to the bottom of one of the peripheral columns **420***a*, **420***b*, **420***c* through a device that can be closed to keep the marine platform **401** on site while the ocean monitoring platform **410** is monitoring the ocean, or that can be open if the ocean monitoring platform **410** needs to be towed back to shore for maintenance or decommissioning. Alternatively, at least one of the peripheral columns **420***a*, **420***b*, **420***c* comprises a short section of chain or rope that is attached to the bottom of the peripheral columns **420***a*, **420***b*, **420***c*. The mooring line or mooring system can be connected to the chain or rope to keep the marine platform **401** on station.
(44) The marine platform **401** comprises a mast **430** on the central column **402** to host a series of equipment and instrumentation, a hatch **431** to enter the central column **402** and a boat landing **432** to access the marine platform **401** with a ladder to climb onboard. The mast **430** is provided for communication and equipment **412** needing to be high or in the open (lidar, bird radar), antennas, etc.
(45) In this exemplary embodiment the marine platform **401** comprises wind turbines **433** on the peripheral columns **420***a*, **420***b*, **420***c* and solar panels **434** that are arranged at the mast **430** on the central column **402**. The wind turbines **433** and the solar panels **434** are electrically connected to not shown batteries. By using a combination of the wind turbines **433**, the solar panels **434** and the batteries the marine platform **401** has zero-emission. In some embodiments, the total power need is equivalent to the capacity of roughly one wind turbine **433** (factoring the site capacity factor). The solar panels **434** may be sized to a minimum power requirement when long periods of low wind speed occur, and some instruments are powered down.
(46) The central column **402** may have four not shown main compartments: a ballast compartment, at the base of the central column **402**, to maintain the expected operational draft and improve the platform stability by lowering its center of gravity, and increasing its metacentric height; a battery storage area, low again for weight control, and vented (such as with a pipe through the top of the center column) to ensure hydrogen or other gaseous formation do not accumulate; a server room, where all the instruments and data boards are racked and interface with a platform server. The server performs the aggregation of the various signals from all instruments, assembles them, performs post analysis as required and transmits to shore the information needed; a storage area for tools and HS&E equipment for visitors or maintenance technicians.
(47) The data measurement and acquisition equipment **412** and therewith the marine platform **401** may be configured to characterize the ocean in very distinct areas. Metocean includes waves (such as using a wave radar for surface mapping and accurate directionality), wind using anemometers and a lidar, current using a submerged ADCP, humidity, air and water temperature, barometric pressure, using specific instruments. Ocean chemistry includes marine growth and corrosion which can be monitored using visual measurements taken over the deployment or the mission of the marine platform **401** on specific plates and cables that can hang from the marine platform **401**. Additionally, salinity, pH and other chemical composition can be measured directly. Biodiversity, wherein the marine platform **401** can be operating a birds and bats radar and can have underwater acoustics to monitor marine mammal migrations. Biohuts may be placed on the marine platform **401** and juvenile fish growth may be measured using both manual diver techniques and acoustics. Similarly, the presence and density of coastal pelagic fish population around the marine platform **401** can be assessed. Ocean-farming, wherein the potential to share a leased site with local fishermen or ocean farmers can have strong benefits, but knowledge of the site is important. Nutrients measurements may be performed as well as the monitoring of growth of various shellfish

and algae. Communication, wherein the marine platform **401** may be fitted with peer-to-peer (P2P) or other communication equipment and may be “connected”. It may provide WiFi locally and possibly cellular signal to the site.

(48) It is to be understood that the above description is included to illustrate the operation of the preferred embodiments and is not meant to limit the scope of the invention. From the above discussion, many variations will be apparent to one skilled in the art that would yet be encompassed by the scope of the present invention.

Claims

1. A floating marine platform comprising: a central column; at least three peripheral columns circumferentially around the central column; and beams radially extending from the central column that connect the peripheral columns with the central column, wherein the beams are steel hollow cylindrical pipes, wherein the peripheral columns comprise a buoyancy air chamber, and wherein the marine platform comprises a motion control system comprising a pressurized air tank discharging air in the buoyancy air chamber, which controls airflow in and out of the buoyancy air chambers using actuated valves and is controlled by a computer system that is coupled with motion sensors, and wherein the pressurized air tank is formed by an airtight inside volume of one of the beams, wherein the floating marine platform further comprises structural members spanning between each adjacent pair of peripheral columns, wherein the structural members are pre-tensioned and comprise or are formed with a steel tube, wherein the peripheral columns comprise a connector having a passage for one end of each of the structural members, and wherein each of the structural members comprises a tension head at an end of the steel tube that is received in the connector, and the steel tube of each of the structural members extends through the passage as from the tension head.
2. The floating marine platform according to claim 1, wherein the peripheral columns comprise a base and wherein the buoyancy air chamber is open to the sea at the base.
3. The floating marine platform according to claim 2, wherein the motion control system comprises an air compressor that is configured to fill the pressurized air tank.
4. The floating marine platform according to claim 3, wherein the motion control system comprises an inlet valve that is configured to control filling of the pressurized air tank by the air compressor.
5. The floating marine platform according to claim 1, wherein the motion control system comprises for each open buoyancy air chamber an outlet valve that is configured to control discharging of air from the pressurized air tank into a corresponding open buoyancy air chamber.
6. The floating marine platform according to claim 1, wherein the motion control system comprises for each open buoyancy air chamber a release valve that connects the buoyancy air chamber to atmosphere and that is configured to control the releasing of air from the open buoyancy air chamber to the atmosphere.
7. The floating marine platform according to claim 1, wherein the peripheral columns and the structural members form a generally triangular shape.
8. The floating marine platform according to claim 1, wherein the structural members are pre-tensioned by inducing a pre-tension stroke thereto that is between 0.04% and 0.07% of a length of each of the structural members.
9. The floating marine platform according to claim 8, wherein the structural members are pre-tensioned by inducing a pre-tension stroke thereto that is 0.05% of a length of each of the structural members.
10. The floating marine platform according to claim 1, wherein the beams are biased in their elongated direction towards the central column.
11. The floating marine platform according to claim 1, wherein the peripheral columns comprise one or more shims or shim plates between the connector and the tension head.

12. The floating marine platform according to claim 11, wherein each of the structural members comprises a tension head on each end of the steel tube, and the peripheral columns comprise two connectors that correspond with each of the tension heads respectively.
13. A floating marine platform comprising: a central column; at least three peripheral columns circumferentially around the central column; and beams radially extending from the central column that connect the peripheral columns with the central column, wherein the beams are steel hollow cylindrical pipes, wherein the peripheral columns comprise a buoyancy air chamber, and wherein the marine platform comprises a motion control system comprising a pressurized air tank discharging air in the buoyancy air chamber, which controls airflow in and out of the buoyancy air chambers using actuated valves and is controlled by a computer system that is coupled with motion sensors, wherein the pressurized air tank is formed by an airtight inside volume of one of the beams, wherein the floating marine platform further comprises structural members spanning between each adjacent pair of peripheral columns, wherein the structural members are pre-tensioned and comprise or are formed with a steel tube, and wherein each of the structural members that extend in a common horizontal plane have a same pre-tension.
14. The floating marine platform according to claim 1, wherein the radially extending beams comprise a top beam and a bottom beam that extend parallel to each other.
15. The floating marine platform according to claim 1, wherein the beams have a circular cross section.
16. The floating marine platform according to claim 1, wherein the central column extends vertically.
17. The floating marine platform according to claim 1, wherein the peripheral columns extend vertically.
18. The floating marine platform according to claim 1, wherein each of the structural members that extend in a common horizontal plane have a same pre-tension.